COMPAISS EVIDENCE BASE

Hallucination Rates in Contemporary AI Systems (2023–2026)

Verified sources only: peer-reviewed studies, official benchmarks, and institutional research reports
Prepared for institutional evaluation | All sources verified June 2026 | Updated edition

50–83%	17–34%	22–94%
Adversarial hallucination rate across 6 major models in clinical vignette study (Mount Sinai / Nature Medicine, 2025)	Hallucination rate in specialized legal RAG tools marketed as 'hallucination-free' (Stanford HAI / RegLab, 2024)	Hallucination/error rates across 26 frontier models on Stanford HAI 2026 factuality benchmark (NEW)

The central finding across all evidence reviewed: Hallucination is not a temporary limitation that improving models will eliminate. It is a persistent, structural property of inference-first AI systems that changes shape as models improve. On tightly constrained, grounded tasks, rates decline. On complex reasoning, open-domain questioning, and adversarial inputs, rates remain high — and in some cases increase as models become more capable. The 2026 Stanford AI Index independently corroborates this conclusion.

VERIFIED EVIDENCE BASE — STUDY BY STUDY

Omar et al. (2025) — Adversarial hallucination attacks in clinical decision support

CLINICAL / NATURE MEDICINE

MODELS TESTED	DEFINITION AND METHOD	REPORTED RATES	SOURCE
GPT-4o, Distilled-DeepSeek-Llama, Phi-4, Gemma-2-27B-it, Qwen-2.5-72B, Llama-3.3-	300 physician-validated vignettes each containing one fabricated clinical detail (lab value, sign, or condition).	Overall default: 65.9% DeepSeek-Llama: 80–83% GPT-4o (best case): 50–53% With mitigation prompt: 44.2%	Peer-reviewed in Communications Medicine (Nature Publishing Group) PMCID: PMC12318031 pmc.ncbi.nlm.nih.gov/articles/PMC12318031 Source verified April 2026

70B (6 models, 5,400 outputs)	Hallucination = model elaborates on fabricated detail as real.	GPT-4o + mitigation: 23%	

Magesh et al. (2024) — Hallucination in specialized legal AI RAG tools

LEGAL / STANFORD HAI

MODELS TESTED	DEFINITION AND METHOD	REPORTED RATES	SOURCE
Lexis+ AI (LexisNexis), Westlaw AI-Assisted Research (Thomson Reuters), Ask Practical Law AI (Thomson Reuters), GPT-4	200+ manually constructed open-ended legal queries. Hallucination = incorrect legal statement or misgrounded citation that does not support the claim made.	Westlaw AI-Assisted Research: >34% Lexis+ AI: >17% Ask Practical Law AI: >17% GPT-4 general-purpose (prior study): 58–82%	Stanford RegLab and HAI hai.stanford.edu/news/ai-trial-legal-models-hallucinate-1-out-6-or-more-benchmarking-queries Source verified April 2026

Key finding: even purpose-built legal RAG tools with 'hallucination-free' marketing claims hallucinated on more than 1 in 6 queries. RAG reduces errors but does not eliminate them.

Graffius (2026) — Are AI Hallucinations Getting Better or Worse?

CROSS-MODEL / BENCHMARK SYNTHESIS

MODELS TESTED	DEFINITION AND METHOD	REPORTED RATES	SOURCE
OpenAI o3, o1; Gemini-2.0-Flash; multiple vendors. Benchmarks: Vectara HHEM, SimpleQA, PersonQA, domain-specific evaluations.	Synthesis of 2024–2025 benchmark data. Two divergent patterns identified: grounded summarization tasks (rates declining) vs. complex open-domain reasoning (rates increasing).	Grounded summarization, top models: 0.7–1.5% OpenAI o3 on PersonQA / SimpleQA: 33–51% OpenAI o1 (prior generation): ~16% Broad task evaluations 2025: 3–20%+	DOI: 10.13140/RG.2.2.33179.53285 scottgraffius.com/blog/files/ai-hallucinations-2026.html Source verified April 2026

Key finding: newer reasoning-focused models hallucinate more on complex factual tasks than earlier models. Stronger reasoning ability does not translate to stronger factual reliability.

Chelli et al. (2024) — Hallucination rates in AI-generated academic citations

CLINICAL / JMIR

MODELS TESTED	DEFINITION AND METHOD	REPORTED RATES	SOURCE
GPT-3.5 (ChatGPT), GPT-4, Bard/Gemini	Generated reference not matching any real paper metadata. Systematic review of AI-generated citations for medical literature.	GPT-3.5: 39.6% GPT-4: 28.6% Bard/Gemini: 91.4%	Journal of Medical Internet Research, 2024, e53164 jmir.org/2024/1/e53164

Vectara HHEM Leaderboard (2025) — Grounded summarization benchmark

BENCHMARK

MODELS TESTED	DEFINITION AND METHOD	REPORTED RATES	SOURCE
Gemini-2.0-Flash, GPT-4o, GPT-o1, DeepSeek-R1, DeepSeek-V3, and others	Hughes Hallucination Evaluation Model (HHEM): faithfulness of model output to a provided source document. Tightly constrained grounded summarization — best-case conditions.	Gemini-2.0-Flash: ~0.7% GPT-o1 / GPT-4o: ~0.9–1.9% DeepSeek-R1: 14.3% DeepSeek-V3: 3.5%	Vectara Hallucination Leaderboard github.com/vectara/hallucination-leaderboard

Note: these are best-case rates on a constrained summarization task — not representative of open-domain or complex reasoning performance.

WHAT THE EVIDENCE ESTABLISHES

The research record across 2023–2025 establishes three findings that are directly relevant to institutional AI governance.

First: hallucination rates are task-dependent, not model-dependent.

The same model family that achieves below 1% on a constrained document summarization task can produce errors on 33–51% of open-domain factual queries. Vectara's leaderboard shows top models approaching 0.7% on grounded summarization. OpenAI's own system card for o3 documents 33–51% error rates on SimpleQA and PersonQA. These are not contradictory findings — they reflect the same underlying architecture operating under different conditions. Institutions evaluating AI on vendor-supplied benchmarks are typically seeing best-case constrained performance, not performance on the kinds of complex, policy-specific questions their users will actually ask.

Second: RAG reduces hallucination but does not eliminate it.

Stanford HAI's independent evaluation of the leading legal AI platforms — tools marketed as 'hallucination-free' by LexisNexis and Thomson Reuters — found hallucination rates of 17–34% in real-world legal query conditions. These are purpose-built RAG systems with curated legal databases. The finding is not that RAG fails — it reduces errors substantially compared to general-purpose models — but that it introduces no architectural guarantee. Hallucination rates depend on retrieval quality, query complexity, and the presence of false premises, all of which vary in real institutional environments.

Third: more capable models can hallucinate more on certain tasks.

The Graffius (2026) synthesis documents a counterintuitive pattern: OpenAI's o3 series, optimized for complex reasoning, hallucinates at 33–51% on open-domain factual recall — more than double the rate of the earlier o1 series at approximately 16%. The Mount Sinai clinical study found that prompt mitigation reduced GPT-4o's adversarial hallucination rate from 53% to 23% — a significant improvement, but still a 23% failure rate under the best available mitigation strategy. Hallucination is not a problem of model quality that will be solved by the next generation. It is a structural property of inference-first architectures that scales with model capability in complex reasoning contexts.

GOVERNANCE IMPLICATION

For regulated institutions, the relevant question is not whether current AI systems hallucinate at acceptable rates on controlled benchmarks. Several do. The question is whether the conditions under which acceptable rates are achieved can be reliably maintained in institutional deployment — and what the consequences are when they are not.

The Stanford HAI legal study is the clearest illustration of this gap. Platforms that claimed to be 'hallucination-free' based on narrow citation accuracy metrics produced incorrect legal information on more than one in six real-world queries. A university, hospital, or government agency that relies on vendor benchmark claims to satisfy its governance obligations is relying on best-case performance figures that may not hold under actual conditions of institutional use.

An architecture that conditions the existence of inference on pre-execution authorization does not compete with these systems on hallucination rate. It removes the structural conditions under which hallucination occurs for unauthorized or out-of-scope queries — not by achieving a lower rate, but by preventing inference from running at all when authoritative sources do not exist to support it.

2026 UPDATE: SUBSEQUENT RESEARCH FINDINGS

Since the initial publication of this evidence base, additional research has emerged that independently corroborates its central claims. This section does not revise the original findings. It documents subsequent evidence that strengthens them.

Stanford AI Index 2026

The Stanford Institute for Human-Centered Artificial Intelligence (HAI) released findings in the 2026 AI Index documenting hallucination and error rates ranging from 22% to 94% across 26 leading AI models on a newly introduced factuality benchmark. Stanford emphasizes that performance varies dramatically by benchmark design and that responsible AI evaluation continues to lag behind capability evaluation.

These findings reinforce a central conclusion of this report: hallucination is not a solved problem, and reported rates are highly dependent on task design, evaluation methodology, and deployment context. Reported hallucination percentages should be interpreted as benchmark-specific measurements, not as universal properties of a model or model family.

Stanford HAI Finding	COMPAiSS Relevance
22%–94% hallucination/error rates across 26 frontier models on new factuality benchmark	Corroborates benchmark-dependence argument. Confirms high rates persist on complex tasks even for leading models.
Models perform substantially worse when false statements are framed as beliefs held by the user vs. a third party	Directly validates COMPAiSS guardrail testing around user pressure, authority framing, and conversational carry-over in Service Canada and Dalhousie scenarios.
Responsible AI evaluation continues to lag behind capability evaluation	Reinforces the governance gap identified in this report: vendors can truthfully report low rates on constrained benchmarks while deployment risk remains high.

Benchmark-Specific Hallucination Rates

The most important governance implication of recent research is that hallucination rates are properties of a model under specific conditions, not intrinsic properties of the model itself. This distinction matters directly for institutional procurement and governance decisions.

The contrast is now well-documented across independent sources. For grounded summarization tasks — where a model is constrained to a provided document — leading systems achieve rates in the low single digits, often 1.8–6% for top-performing systems on 2026 leaderboards. For open-ended factual generation — closer to policy drafting, advisory work, and knowledge-work use cases — the 2026 Stanford AI Index factuality benchmark finds hallucination and error rates between 22% and 94% across 26 frontier models. These are the same models, the same vendors, and in some cases the same deployment versions.

For public-sector readers: a vendor can truthfully report a low hallucination rate while their system remains high-risk in the deployment context that actually matters. The benchmark conditions under which a low rate is achieved are rarely the conditions of institutional use.

User Framing and Sycophancy Effects

Multiple independent sources now document that models are more likely to produce inaccurate outputs when prompted to validate or elaborate beliefs attributed directly to the user. The 2026 Stanford AI Index reports that one widely deployed frontier system fell from over 90% accuracy to approximately 14% when identical false statements were framed as user-held beliefs rather than third-party beliefs — a collapse in accuracy attributable entirely to framing, with no change in the underlying factual content of the query.

Related analyses of sycophancy-inducing prompts report hallucination rates above 50% for general-purpose models when users express strong prior beliefs or authority cues. These behavioral patterns closely mirror COMPAiSS guardrail tests conducted in Service Canada and Dalhousie scenarios involving user pressure, conversational framing, authority assumptions, retrieval-failure substitution, and context carry-over. Independent academic benchmarking and governance-oriented testing are converging on the same failure modes.

For public-sector deployments: interaction styles that are user-deferential or 'customer-is-always-right' in orientation can substantially increase hallucination risk when users express strong prior beliefs or assert authority. This is not a fringe edge case. Stanford's 2026 data shows accuracy drops of 80+ percentage points under user-belief framing for some frontier models.

Structural Nature of Hallucination

Formal theoretical work published since 2024 has strengthened the conclusion that hallucination is a structural feature of current large language models rather than a minor engineering defect. Several researchers have argued that hallucinations may be an inherent property of probabilistic generative systems operating under uncertainty, suggesting that hallucination risk can be reduced but may not be completely eliminated under current architectural approaches.

A synthesis of approximately thirty empirical studies published between 2023 and 2026 similarly concludes that hallucinations arise from the probabilistic nature of model generation and the mismatch between training distributions and deployment contexts. These findings corroborate the conclusion originally advanced in this report: hallucination should be treated as a persistent governance and safety concern rather than a short-term product defect that will be resolved by the next model generation.

Evaluation Variability and Prompt Sensitivity

Emerging research on evaluation design demonstrates that hallucination measurements are themselves sensitive to prompt phrasing and evaluation setup — not merely to task type. A 2026 study published in the proceedings of the European Chapter of the Association for Computational Linguistics (EACL) introduces a prompt multiplicity framework and reports over 50% inconsistency in model outputs across different but semantically equivalent prompts on medical hallucination benchmarks.

This finding has a direct implication for governance: single-headline hallucination percentages are not sufficient for institutional risk assessment, regardless of their source. Robust evaluation requires diverse prompts, scenarios, and interaction styles. It also validates the COMPAiSS decision to test governance risk through multiple scenario types rather than a single synthetic benchmark — a methodology that independent research now confirms is necessary rather than merely cautious.

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